

CS 5200 Database Management Systems

Lab 3: Setting Up MySQL Database on GCP

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Setting Up MySQL Database on your GCP

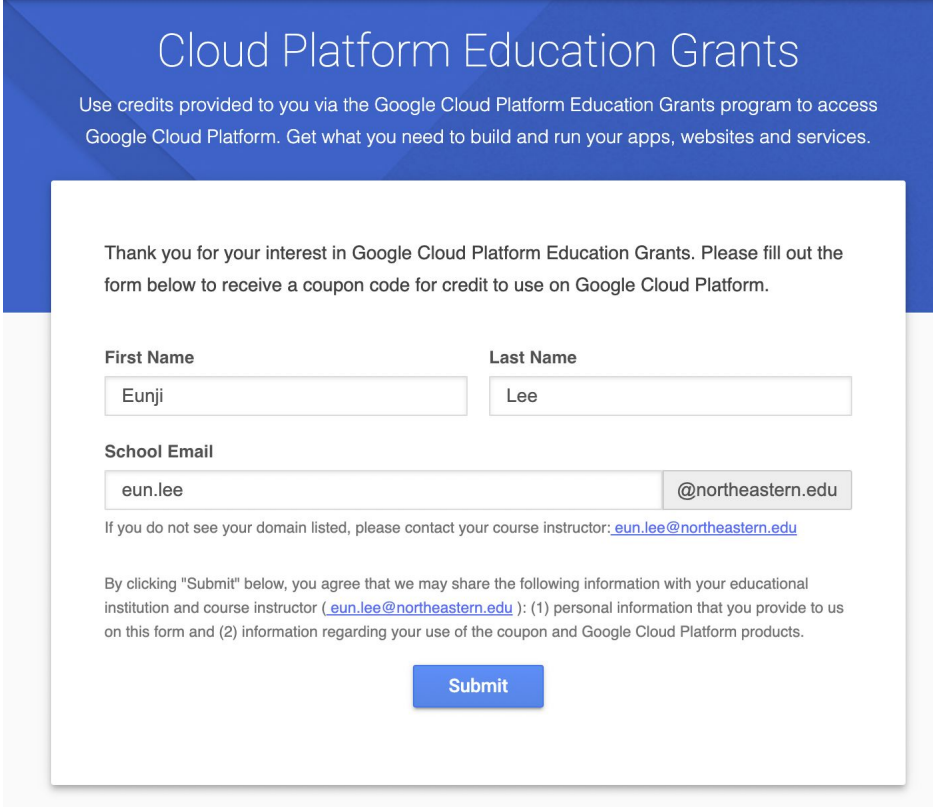
- There 3 different ways to install MySQL database on your GCP Compute Engine as below:
 1. **Google Cloud SQL**
 2. Google Cloud Launcher
 3. MySQL on Compute Engine
- In this exercise lab, we are going to use Option 1 to create MySQL instance. However, students can create MySQL instance using either Option 2 or 3.

A. Activate a GCP coupon

Here is the URL you will need to access in order to request a Google Cloud coupon. You will be asked to provide your school email address and name. An email will be sent to you to confirm these details before a coupon is sent to you. **To redeem the coupon itself, use your NEU email.**

[Student Coupon Retrieval Link](#)

- You will be asked for a name and email address, which needs to match your school domain. A confirmation email will be sent to you with a coupon code.
- You can request a coupon from the URL and redeem it until: **1/5/2026**
- Coupon valid through: **9/5/2026**
- You can only request ONE code per unique email address.



The screenshot shows the 'Cloud Platform Education Grants' form. At the top, it says 'Use credits provided to you via the Google Cloud Platform Education Grants program to access Google Cloud Platform. Get what you need to build and run your apps, websites and services.' Below this, a message says 'Thank you for your interest in Google Cloud Platform Education Grants. Please fill out the form below to receive a coupon code for credit to use on Google Cloud Platform.' The form has three input fields: 'First Name' with 'Eunji', 'Last Name' with 'Lee', and 'School Email' with 'eun.lee' and '@northeastern.edu'. Below the email field, it says 'If you do not see your domain listed, please contact your course instructor: eun.lee@northeastern.edu'. At the bottom, there is a 'Submit' button and a disclaimer: 'By clicking "Submit" below, you agree that we may share the following information with your educational institution and course instructor (eun.lee@northeastern.edu): (1) personal information that you provide to us on this form and (2) information regarding your use of the coupon and Google Cloud Platform products.'

A. Activate a GCP coupon

Once you get the credit, you will get an email with your unique coupon code and a link to activate it

1. When you redeem the coupon, you will need to provide an [@gmail.com](#) address. Please follow prompts to create one if you do not have a personal Gmail address.

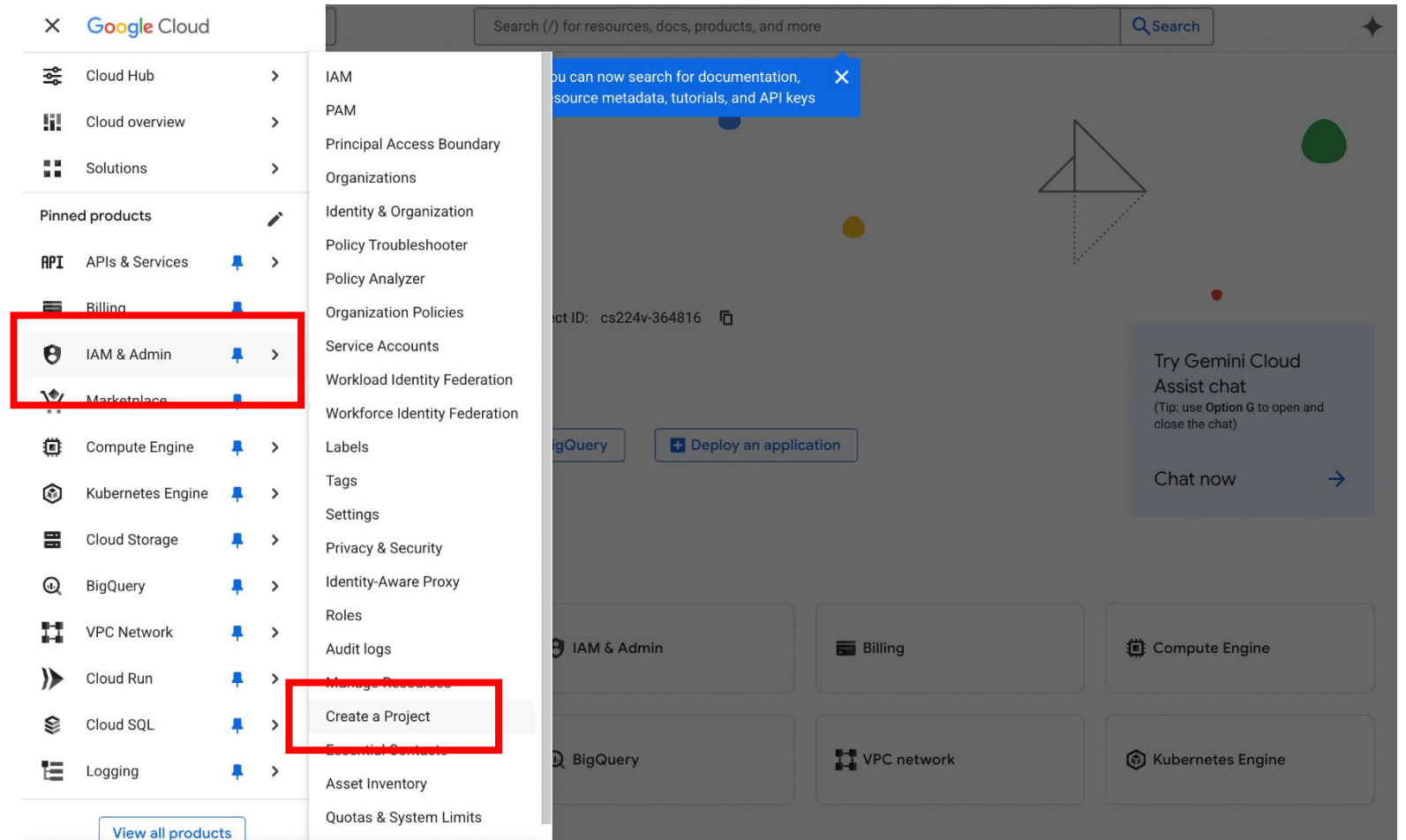
B. Create a Project

Create a project

- Go to console.cloud.google.com
- You create your first project using the Google Cloud Platform Console. The project is used to complete all of the labs in this course.

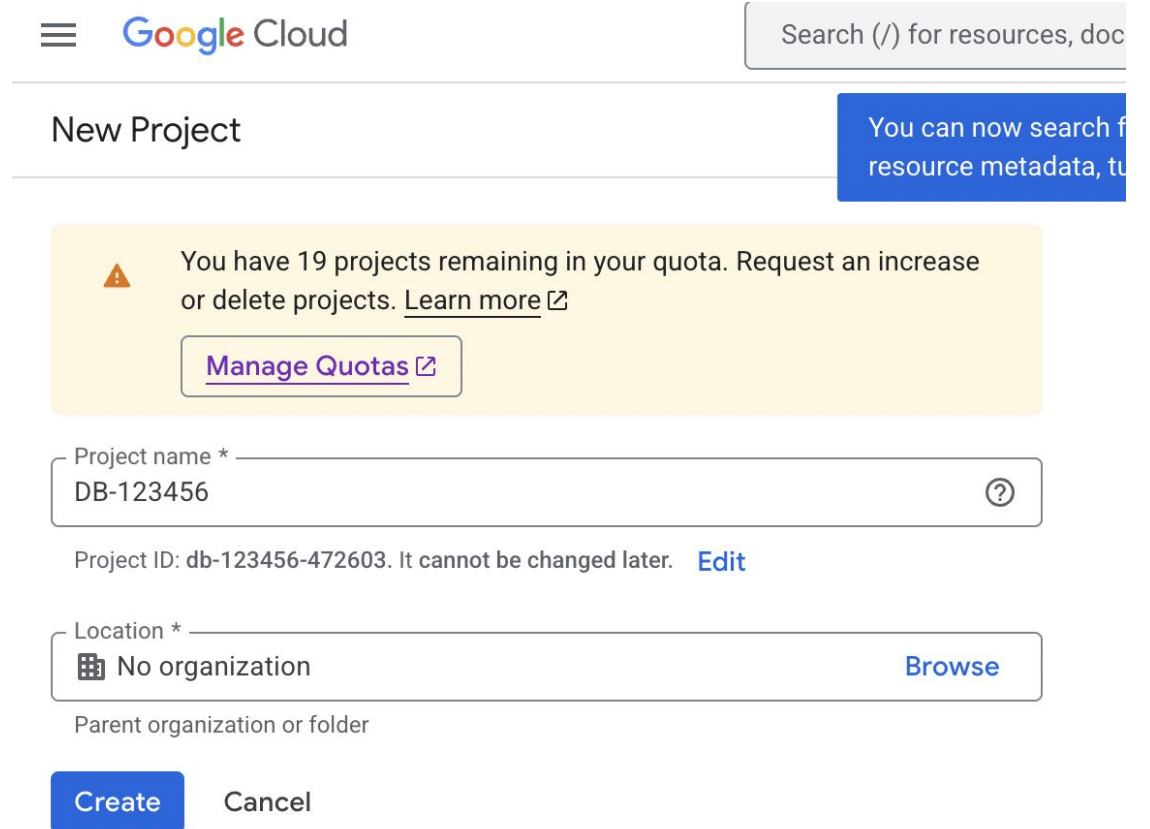
To create a project:

- **Step 1:** In the top left corner of the Google Cloud Platform Console, click the 3-line icon.
- Navigate to click **IAM & Admin > Create a project**.



Step 2: In the 'New Project' dialog:

- For **Project name** type **[Subject Name] + "-" + [NEU ID]**. For example:
DB-123456
- Copy down the **Project ID** in the text below the project name box; you need it later.
- For location, use "No Organization"
- For **Please email me updates** choose an appropriate option. This option only appears the first time you create a project for a newly registered account. If it does not appear, proceed to the next option.
- If asked, click **Yes** to agree to the terms of service.
- Click **Create**.



Google Cloud

Search (/) for resources, doc

New Project

You can now search for resource metadata, tu

 You have 19 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[Manage Quotas](#)

Project name *
DB-123456

Project ID: db-123456-472603. It cannot be changed later. [Edit](#)

Location *
 No organization [Browse](#)

Parent organization or folder

[Create](#) Cancel

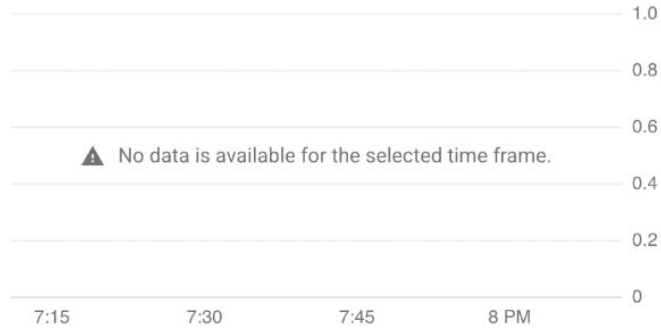
you can now search for documentation,
source metadata, tutorials, and API keys



API APIs



Requests (requests/sec)



→ Go to APIs overview

Notifications

Customize



Create Project: DB-123456

Just now

Select Project

See all activities



Go to Cloud status dashboard



Billing



Estimated charges
For the billing period Sep 1 – 18, 2025

USD \$0.00



Take a tour of billing



View detailed charges



Monitoring



Create my dashboard
Set up alerting policies
Create uptime checks

View all dashboards



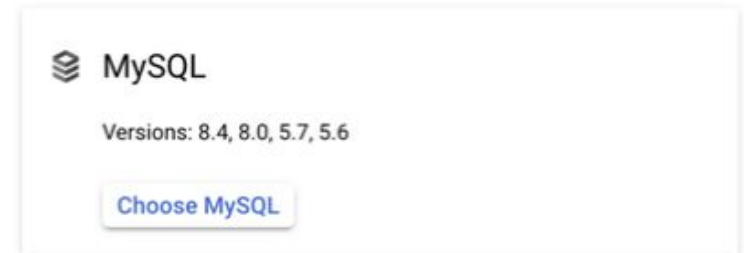
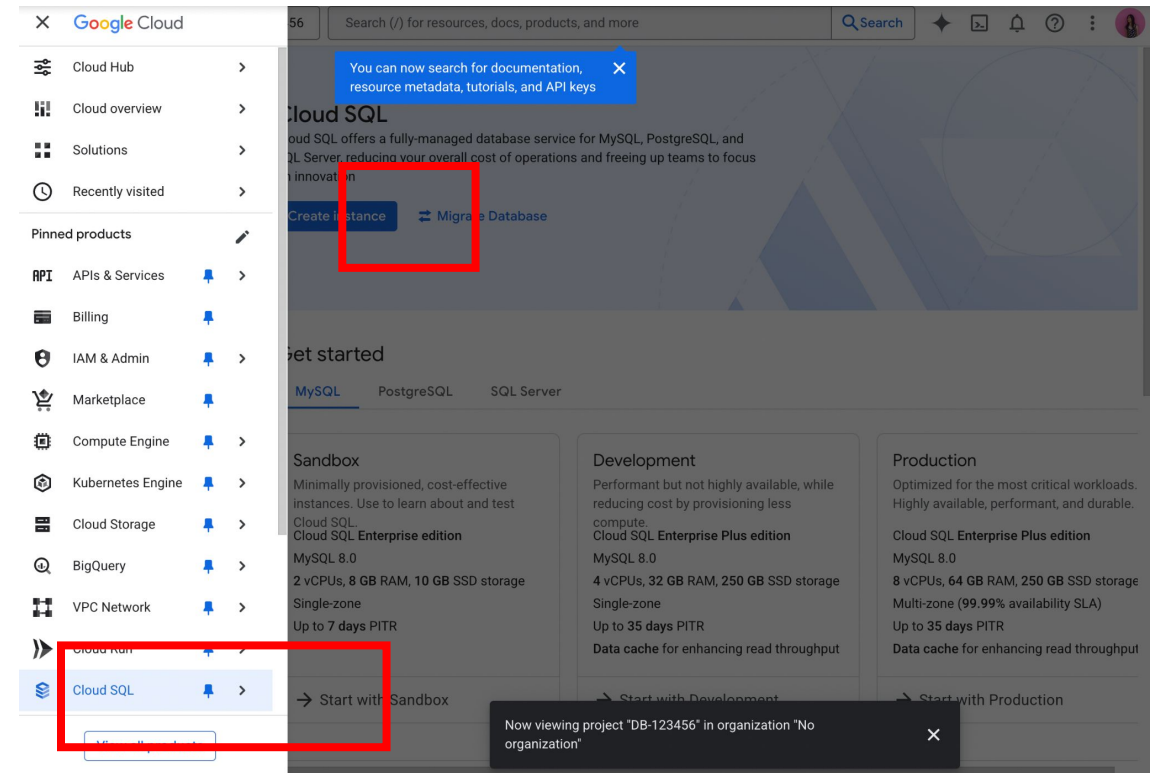
Go to Monitoring

C. Creating MySQL instance

1. Go to the Cloud SQL Instances page in the Google Cloud Platform Console.

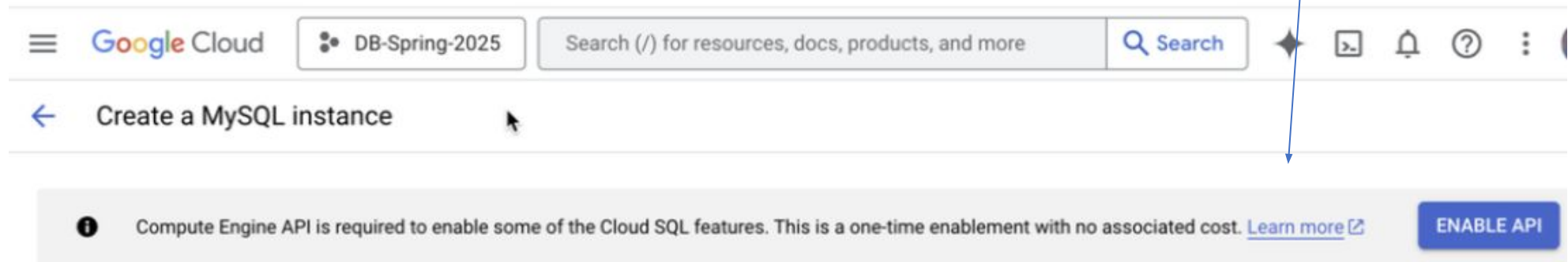
GCP Console -> Cloud SQL

2. Then, click **“Create Instance”**
3. Select **MySQL** and click **Next**.
4. Click **Choose MySQL**.



If you see the following...

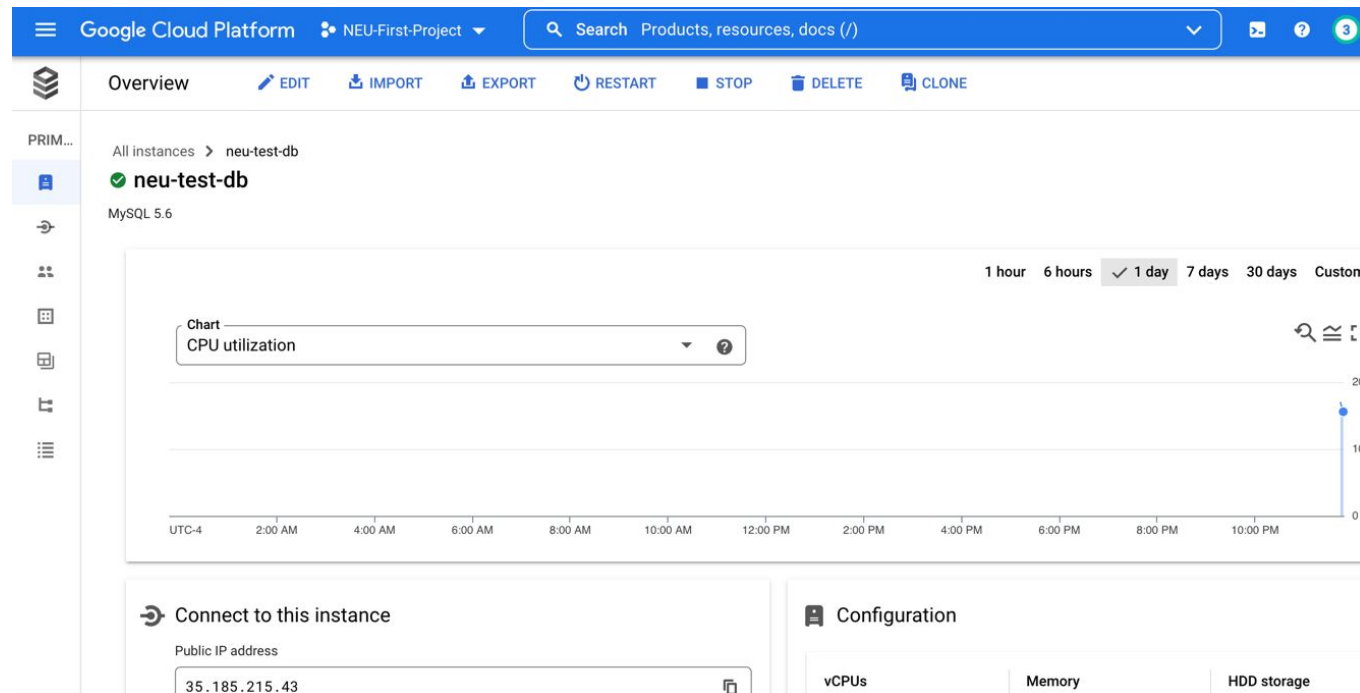
Click “ENABLE API”



1. Choose a Cloud SQL edition: **“Enterprise”**
2. Choose an edition preset for this edition: **“Sandbox”**
3. Enter the following info:
 - Instance ID: **neu-test-db**
 - Root password: **Must setup password** (please make one for yourself and note it down so you remember!)
 - Choose database version: **MySQL 8.0**
 - Region: **us-west1 (Oregon)**
 - Zone availability: **Single zone**
 - Primary zone: Any
4. Show **“Customize your instance”** option to change any of the following options:
 - Machine Type: **Shared core (1 vCPU, 0.614 GB)**
 - Storage type: **HDD**
 - Storage capacity: **10 GB**
 - **Enable automatic storage increase**
 - Set connectivity: check **Public IP**
 - Add a network:
 - Name: **All IP**
 - Network: **0.0.0.0/0**
 - Then, click **“Done”**
 - Backups: **uncheck automatic backups**
5. Click **Create (It will take approximately 10 ~ 15 minutes)**

D. Maintaining MySQL instance

- Viewing information about MySQL instance
 - Select an instance to view by clicking the instance name.
 - Cloud Platform Console displays the Instance details pane for the instance.



Configuration

vCPUs
2

Memory
8 GB

SSD storage
10 GB

 Enterprise edition [Upgrade](#)

 Database version is MySQL 8.0.41

 Auto storage increase is enabled

 Backup Tier: Standard

 Automated backups are disabled

 Point-in-time recovery is disabled

 Instance deletion prevention is enabled

 Backup retention after deletion is enabled

 Located in us-west1-a

 Not highly available (zonal)

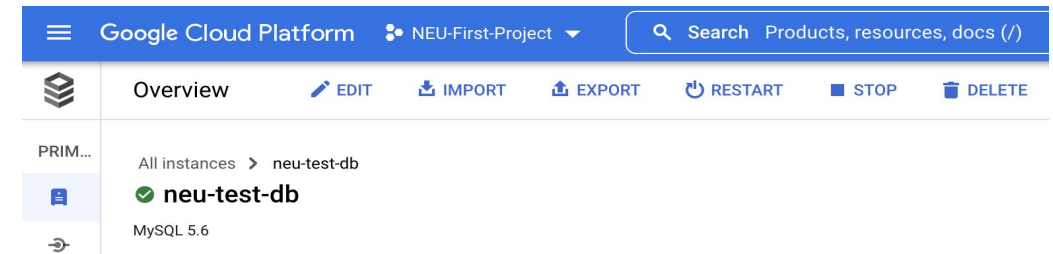
 No database flags set

 No labels set

[→ Edit configuration](#)

Starting, Stopping, Restarting Instance

- Starting an Instance
 - Select an instance to start by clicking the instance name.
 - Click Start, in the button bar at the top of the page.
 - In the Start database instance? dialog box, click Start.
 - The activation policy of the instance is set to Always and the instance is started.
- Stopping an Instance
 - Select an instance to restart by clicking the instance name.
 - Click Stop, in the button bar at the top of the page.
 - In the Stop database instance? dialog box, click Stop.
 - The activation policy of the instance is set to Off (Never) and the instance is stopped.
- Restarting an Instance
 - Select an instance to restart by clicking the instance name.
 - Click Restart, in the button bar at the top of the page.
 - In the Restart database instance dialog box, click Restart.



IMPORTANT! STOP YOUR INSTANCE WHEN YOU ARE NOT USING IT!

Your \$100 GCP credits should last you throughout the semester but *only if you remember to turn off your instance when you are not using it!*

Please just “STOP” and do not “DELETE” the instance!

To check how many credits you have left, go to “Billing Account Management” > “Credits”.

We do not have enough duplicate credits for everyone. If you run out of credits due to forgetting to turn it off, there is nothing the course staff can do.

IMPORTANT! STOP YOUR INSTANCE WHEN YOU ARE NOT USING IT!

The screenshot shows the Google Cloud Billing console. The left sidebar contains a navigation menu with categories: Billing account (Billing Account for Education), Overview, Cost management (Reports, Cost table, Cost breakdown, Budgets & alerts, Billing export, Anomalies), Cost optimization (FinOps hub, Committed use discounts..., CUD analysis, Pricing, Cost estimation, Credits), and Billing management (Account management). The 'Credits' menu item is highlighted with a red box. The main content area is titled 'Credits' and shows a table of credit details. The top right corner features a user profile dropdown menu, which is also highlighted with a red box. The dropdown menu includes options: Billing account management, Preferences, Downloads, Cloud Partners, Terms of service, Privacy, and Project settings.

Google Cloud

Search (/) for resources, docs, products, and more

Billing / Credits / All Credits

Billing account
Billing Account for Education

Overview

Cost management

Reports

Cost table

Cost breakdown

Budgets & alerts

Billing export

Anomalies

Cost optimization

FinOps hub

Committed use discounts...

CUD analysis

Pricing

Cost estimation

Credits

Account management

Credits

All Credits

View and download credit details here. Active committed use discounts are not included here and can be viewed on the [Commitments page](#).

Filter Filter credits

Credit name	Status ↑	Percent remaining	Remaining value	Original value	Type	Credit ID	Scope ⓘ
CS 5200: Database Management Systems	Available	100%	\$100.00	\$100.00	One-time	E1J7V4RX0...	Any service on this billing account.

2025 2026

Billing account management

Preferences

Downloads

Cloud Partners

Terms of service

Privacy

Project settings

IMPORTANT! STOP YOUR INSTANCE WHEN YOU ARE NOT USING IT!

The screenshot shows the Google Cloud console interface. At the top, there's a navigation bar with the Google Cloud logo, a project ID 'DB-123456', a search bar, and various utility icons. Below this, the 'Cloud SQL' section is active, showing a sidebar with 'Instances' and 'Backups'. The main content area is titled 'Instances' and includes buttons for 'Create Instance' and 'Migrate Database'. A notification banner at the top of the main area informs about extended support for PostgreSQL and MySQL instances starting in February 2025. Below the notification is a table of instances. The table has columns for Status, Instance ID, Issues, Cloud SQL edition, Type, Public IP address, Private IP address, Instance connection name, High, and Actions. One instance, 'neu-tes-db', is listed with a status of 'Stopped', which is highlighted by a red box. The 'Stopped' label is also visible in a button below the instance name.

Google Cloud DB-123456 Search (/) for resources, docs, products, and more Search

Cloud SQL Instances Create Instance Migrate Database Show info panel Learn

Starting Feb 1, 2025, all instances running community end-of-life versions of PostgreSQL and MySQL are under extended support. These instances will be charged for extended support from May 1, 2025. Upgrade your instances running end-of-life versions before May 1, 2025 to prevent additional charges. [Learn more](#)

View affected instances Dismiss

Filter Enter property name or value

Status	Instance ID	Issues	Cloud SQL edition	Type	Public IP address	Private IP address	Instance connection name	High	Actions
Stopped	neu-tes-db		Enterprise	MySQL 8.0	34.83.211.26		db-123456-472603:us-...	Ena	

Status should show as “stopped” when you are not using it.

[Lab3 Assignment] MySQL on GCP

- Create MySQL instance on GCP, then submit a screenshot of GCP console that shows a created MySQL database instance
- Submit your screenshot to Canvas

Next lab, we will learn how to connect to your MySQL instance and create a database!

What should your submission look like?

Google Cloud DB-123456

Search (/) for resources, docs, products, and more

Cloud SQL Overview Edit Import Export Restart Stop Delete Clone

Primary instance

Overview

All instances > neu-test-db

neu-test-db

MySQL 8.0

Learn the basics of Cloud SQL

0 of 3 completed

- 1 Import data
- 2 Query and explore data
- 3 Connect source application

With your Cloud SQL instance now running, the next step is to import data. You can import CSV and SQL files from your local file system or from a GCS bucket.

If you don't have existing data, you can [create a database](#), then create a table in Cloud SQL Studio in the next step.

Import data Mark as complete

Chart

CPU utilization

1 hour 6 hours 1 day 7 days 30 days Custom

UTC-7 10:00 PM 11:00 PM Sep 18 1:00 AM 2:00 AM 3:00 AM 4:00 AM 5:00 AM 6:00 AM 7:00 AM 8:00 AM 9:00 AM 10:00 AM 11:00 AM 12:00 PM 1:00 PM 2:00 PM 3:00 PM 4:00 PM 5:00 PM 6:00 PM 7:00 PM 8:00 PM

Go to Query insights for more in-depth info on queries and performance

Instance health issues (0)

No issues found

Configuration

2 vCPUs

8 GB Memory

Uploads and DB-123456 operations

Started neu-test-db	8:59:27 PM GMT-7
Stopped neu-test-db	8:49:42 PM GMT-7

Clearly show the project name with your NUID.

Show your Instance running.